Transition Systems and Linear Temporal Logic

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Linear Temporal Logic (LTL)

Introduction to LTL

- LTL can be used to specify important system properties.
- For reactive systems, correctness depends on the *executions* of the system – not only the input and output of a computation – and on fairness issues.
    - Temporal logic is a formalism for treating these aspects.
- LTL extends propositional or predicate logic by modalities that permit to referral to the infinite behavior of a reactive system.

Introduction to LTL

- Provides a very intuitive but mathematically precise notation for expressing properties about the relation between the state labels in execution.
- Imposes temporal modalities on top of predicate logic.
- Although the word "temporal" suggests a relationship with the real-time behavior of a reactive system, this is only true in an abstract sense
    - Allows for the specification of the relative order of events

Introduction to LTL

Syntax of LTL

Definition 5.1. Syntax of LTL

LTL formulae over the set AP of atomic proposition are formed according to the following grammar:[1]

$$\varphi ::= \text{true} \mid a \mid \varphi_1 \wedge \varphi_2 \mid \neg\varphi \mid \bigcirc \varphi \mid \varphi_1 \mathsf{U} \varphi_2$$

where $a \in AP$. ■

Syntax of LTL

$$
\begin{array}{lcl}
\varphi_1 \vee \varphi_2 & \stackrel{\text{def}}{=} & \neg(\neg\varphi_1 \wedge \neg\varphi_2) \\
\varphi_1 \rightarrow \varphi_2 & \stackrel{\text{def}}{=} & \neg\varphi_1 \ \vee \ \varphi_2 \\
\varphi_1 \leftrightarrow \varphi_2 & \stackrel{\text{def}}{=} & (\varphi_1 \rightarrow \varphi_2) \wedge (\varphi_2 \rightarrow \varphi_1) \\
\varphi_1 \oplus \varphi_2 & \stackrel{\text{def}}{=} & (\varphi_1 \wedge \neg\varphi_2) \ \vee \ (\varphi_2 \wedge \neg\varphi_1)
\end{array}
$$

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Syntax of LTL

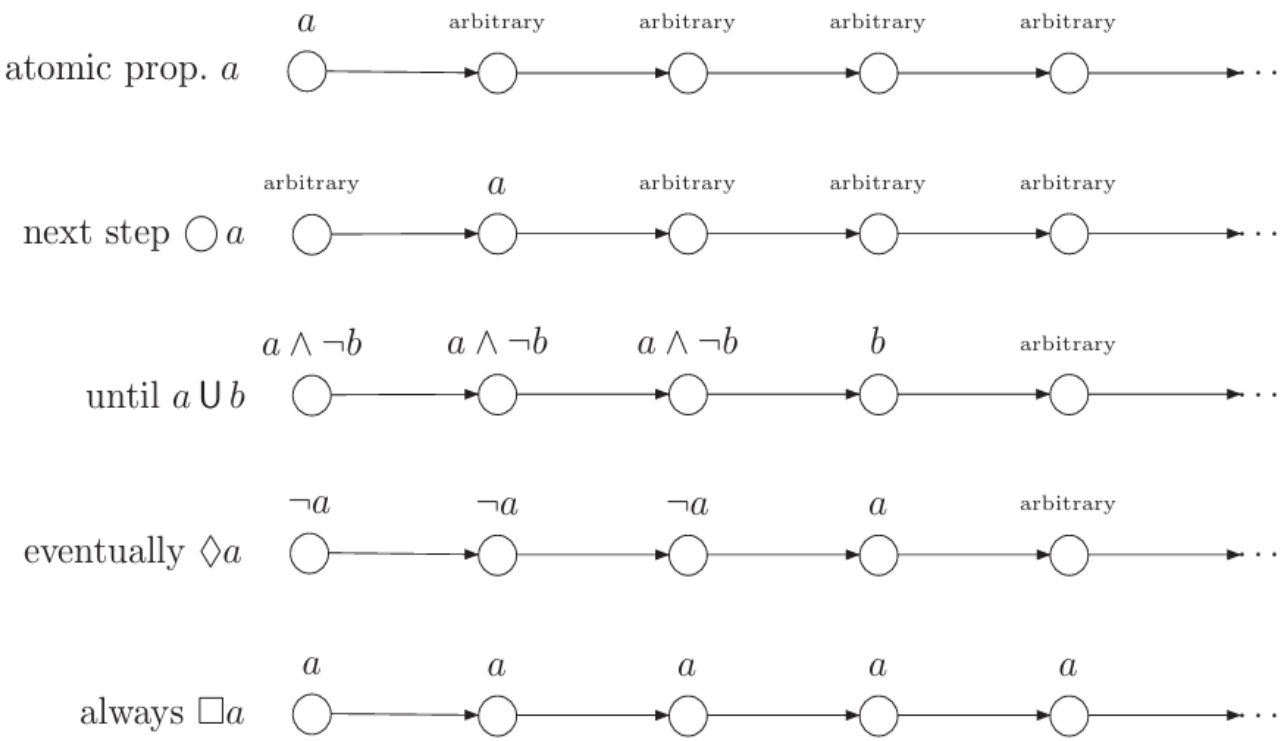


Figure 5.1: Intuitive semantics of temporal modalities.

Examples of LTL